The Cartesian coordinate system

Two number lines at right angles, and every point of the plane named by an ordered pair.

LESSON 137–138 | 2 × 40 MIN | MATEMATIKA 7, §59 | S8 11

LESSON CLOCK

15' | 20' | 25' | 20' | 5'

Two axes and one origin	15 min
Plotting and reading	20 min
The four quadrants	25 min
Lengths and midpoints	20 min
Homework	5 min

LEARNING OBJECTIVES

- Plot a point from its coordinates and read the coordinates of a plotted point.
- Name the four quadrants and the signs in each.
- Recognise the points of the axes by their coordinates.
- Find lengths and midpoints of segments parallel to an axis.

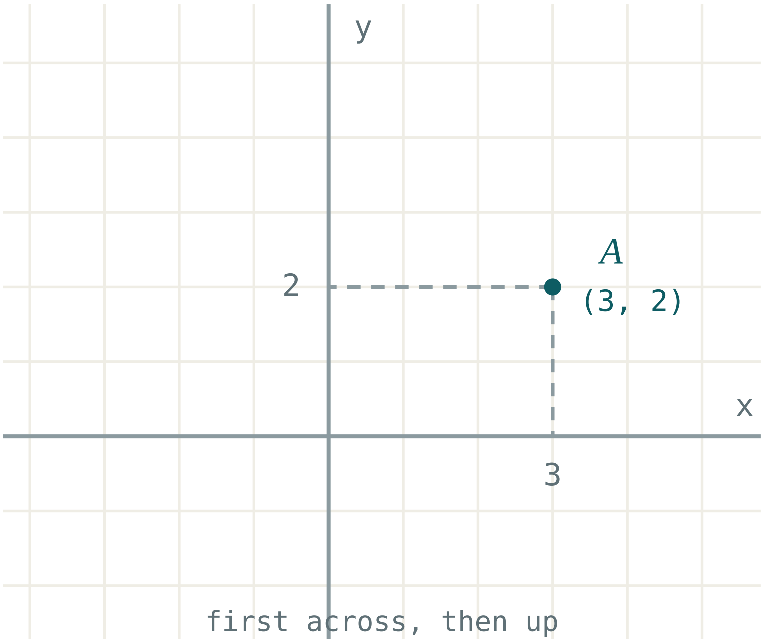
TEXTBOOK

National	pp. 407–414
Cambridge	Stage 8, pp. 106–116
Workbook	Exercise 11.1

01 Explanation

Two axes and one origin

Two perpendicular number lines meet at the **origin** $O(0; 0)$: the horizontal Ox and the vertical Oy. Every point is then named by an ordered pair.



A point and its two coordinates

NAME	WHICH NUMBER	READ FROM
abscissa x	the first	the horizontal axis
ordinate y	the second	the vertical axis

THE ORDER IS FIXED: ALONG FIRST, THEN UP

$A(3; 2)$ and $B(2; 3)$ are different points. “Ordered pair” means exactly this — the pair carries information in its order, not only in its two numbers.

Plotting and reading

POINT	GO ALONG	THEN	WHERE IT LANDS
$A(3; 2)$	3	up 2	quadrant I
$B(-4; 1)$	left 4	up 1	quadrant II
$C(-2; -3)$	left 2	down 3	quadrant III
$D(5; -1)$	5	down 1	quadrant IV
$E(0; 4)$	nowhere	up 4	on the y-axis
$F(6; 0)$	6	nowhere	on the x-axis

A ZERO TELLS YOU WHICH AXIS

$x = 0$ puts the point on the vertical axis; $y = 0$ puts it on the horizontal one. Both zero is the origin itself.

The four quadrants

QUADRANT	WHERE	X	Y	EXAMPLE
I	top right	+	+	$(3; 2)$
II	top left	−	+	$(-4; 1)$
III	bottom left	−	−	$(-2; -3)$
IV	bottom right	+	−	$(5; -1)$

THEY ARE NUMBERED ANTICLOCKWISE

Starting at the top right and turning the way a positive angle turns. The numbering is a convention, but it is the same convention worldwide.

Lengths and midpoints

When a segment is parallel to an axis, its length is the difference of the coordinates that change, and its midpoint is their average.

SEGMENT	LENGTH	MIDPOINT
$A(1; 3)$ to $B(7; 3)$	6	$(4; 3)$
$C(2; -1)$ to $D(2; 5)$	6	$(2; 2)$
$E(-3; 4)$ to $F(4; 4)$	7	$(0.5; 4)$

THE IDEA BEHIND THE COORDINATE PLANE

Descartes' insight was that a geometric question — where is the midpoint? — becomes an arithmetic one: take the average. Grade 8 turns every line and circle into an equation on exactly this principle.

02 Worked examples

EXAMPLE 1 In which quadrant does $(-2; -3)$ lie?

1. x is negative: left of the y-axis.
2. y is negative: below the x-axis.
3. Bottom left — quadrant III.

ANSWER

Quadrant III

EXAMPLE 2 Find the length and the midpoint of the segment from $A(1; 3)$ to $B(7; 3)$.

1. The ordinates are equal, so the segment is horizontal.
2. Length $= 7 - 1 = 6$.
3. Midpoint abscissa $= \dfrac{1 + 7}{2} = 4$.
4. Midpoint $(4; 3)$.

ANSWER

Length 6, midpoint $(4; 3)$

EXAMPLE 3 **Three vertices of a rectangle are $(1; 1)$, $(5; 1)$ and $(5; 4)$. Find the fourth.**

1. The side from $(1; 1)$ to $(5; 1)$ is horizontal.
2. The side from $(5; 1)$ to $(5; 4)$ is vertical.
3. The fourth vertex is above $(1; 1)$ by 3.
4. $(1; 4)$

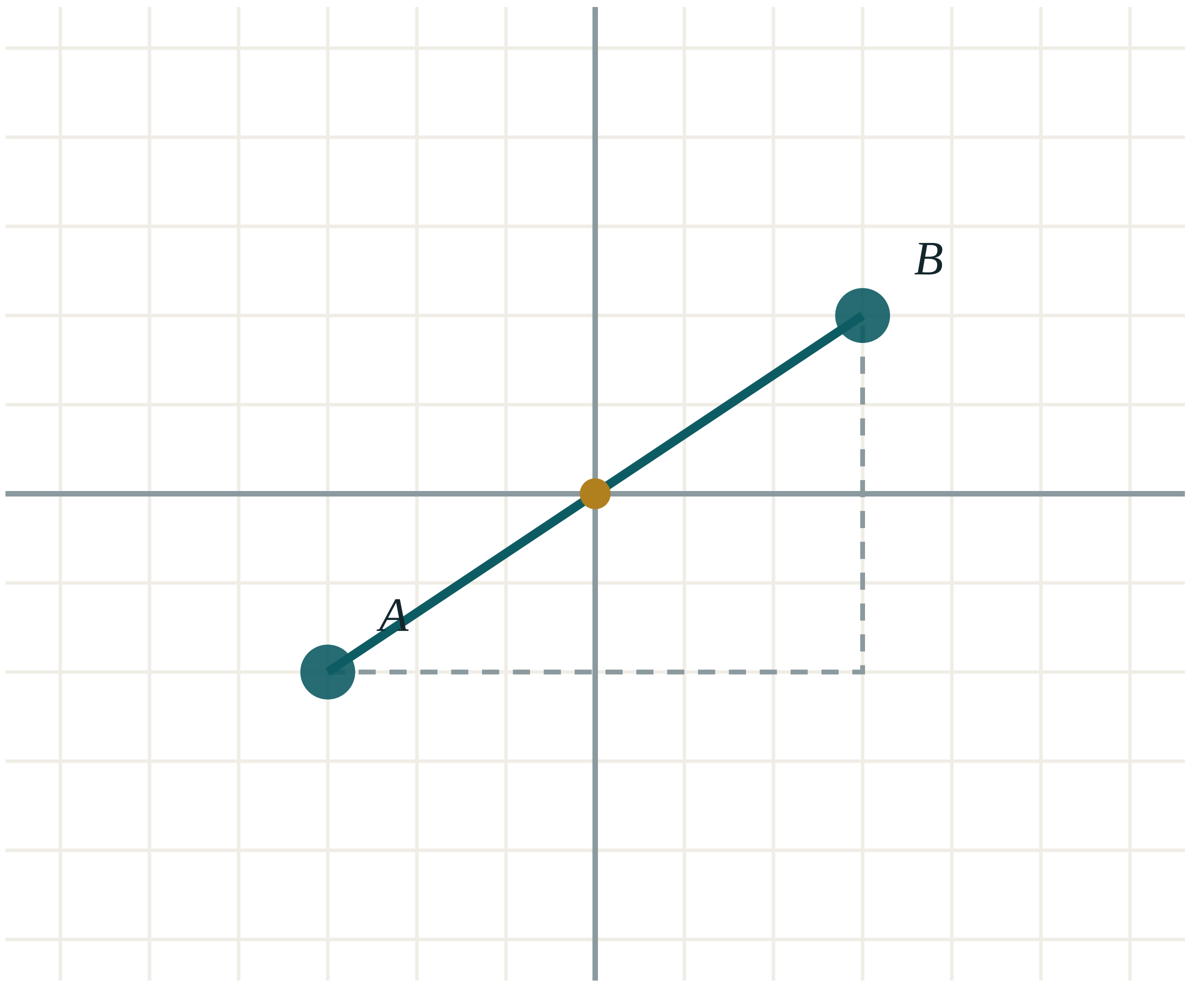
ANSWER

$(1; 4)$

03 Interactive model

Play “battleships” on squared paper for ten minutes with coordinates instead of letters; the order of the pair stops being a rule and becomes a habit.

Two points, their difference and their midpoint



A (-3, -2)	B (3, 2)	$x_2 - x_1$ 6	$y_2 - y_1$ 4	distance AB 7.21	midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Coordinate plane	Koordinata tekisligi	Координатная плоскость
Axis	O‘q	Ось
Origin	Koordinata boshi	Начало координат
Abscissa	Abssissa	Абсцисса
Ordinate	Ordinata	Ордината
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Quadrant	Chorak	Четверть
To plot	Belgilamoq	Отметить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first number of a pair is the:

- ordinate
- abscissa
- origin
- quadrant

(3; 2) and *(2; 3)* are:

- the same point
- different points
- both on an axis
- both in quadrant II

(−4; 1) lies in quadrant:

- I
- II
- III
- IV

$(5; -1)$ lies in quadrant:

- I
- II
- III
- IV

$(0; 4)$ lies:

- in quadrant I
- on the y-axis
- on the x-axis
- at the origin

The midpoint of $(1; 3)$ and $(7; 3)$ is:

- $(4; 3)$
- $(3; 3)$
- $(8; 6)$
- $(4; 6)$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The quadrant of $(3; 2)$

   ANSWER I

2. The quadrant of $(-4; 1)$

   ANSWER II

3. The quadrant of $(-2; -3)$

   ANSWER III

4. The quadrant of $(5; -1)$

   ANSWER IV

5. Where is $(0; 4)$?

   ANSWER On the y-axis

6. Where is $(6; 0)$?

   ANSWER On the x-axis

7. The coordinates of the origin

   ANSWER $(0; 0)$

Medium · the standard question

7 PROBLEMS

1. Length of $A(1; 3)$ to $B(7; 3)$

   ANSWER 6

2. Its midpoint

   ANSWER $(4; 3)$

3. Length of $C(2; -1)$ to $D(2; 5)$

   ANSWER 6

4. Its midpoint

   ANSWER $(2; 2)$

5. The fourth vertex of a rectangle on $(1; 1)$, $(5; 1)$, $(5; 4)$

   ANSWER $(1; 4)$

6. The reflection of $(3; 2)$ in the x-axis

   ANSWER $(3; -2)$

7. The reflection of $(3; 2)$ in the y-axis

   ANSWER $(-3; 2)$

Hard · stretch and reason

7 PROBLEMS

1. The area of the rectangle with vertices $(1; 1)$, $(5; 1)$, $(5; 4)$, $(1; 4)$

   ANSWER 12

2. Its perimeter

   ANSWER 14

3. All points with $y = 3$ form

   ANSWER a horizontal line

4. All points with $x = -2$ form

   ANSWER a vertical line

5. All points with $x = y$ form

   ANSWER the line through the origin bisecting quadrants I and III

6. The midpoint of $(-3; 4)$ and $(4; 4)$

   ANSWER $(0.5; 4)$

7. A point equidistant from both axes in quadrant III

   ANSWER e.g. $(-2; -2)$

07 Homework

Homework — 5 tasks

Along first, then up — and label every point you plot.

1. Plot $A(2; 5)$, $B(-3; 4)$, $C(-1; -2)$ and $D(4; -3)$, and name each quadrant.
2. Find the length and midpoint of the segment from $(-2; 3)$ to $(6; 3)$.
3. Three vertices of a rectangle are $(2; 1)$, $(8; 1)$ and $(8; 5)$. Find the fourth and the area.
4. Where do all the points with $y = -4$ lie?
5. Reflect $(-5; 2)$ in each axis and give both images.